

# CUBIC-FIT: A High Performance and TCP CUBIC Friendly Congestion Control Algorithm

Sudip Kumar Saha  
2105152

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# 1 Introduction

TCP congestion control plays a critical role in ensuring efficient and fair data transmission over networks. While TCP CUBIC performs well in wired environments, it often struggles in wireless networks due to random packet losses, mobility, and interference that are not caused by actual congestion.

This project evaluates CUBIC-FIT, a modified version of it, designed to be more robust in wireless conditions while remaining friendly to standard CUBIC flows. We proposed three key enhancements — Dynamic Alpha adaptation, Bottleneck Queue Estimation, and Simulated Multi-Flow Aggregation — and tested them in both mobile 802.11 wireless and wired dumbbell topologies through extensive parameter sweeps.

The goal was to observe how these modifications improve throughput, packet delivery, and stability in challenging real-world-like scenarios.

## 2 Proposed Modifications

The following modifications were implemented in the NS-3 `TcpCubicFit` module to improve CUBIC performance in wireless environments.

### 2.0.1 Plan 1: Dynamic Alpha ( $\alpha$ ) Adaptation

This plan is derived from Section III of the paper, specifically the CUBIC-FIT friendliness analysis under Theorem 2 using Bandwidth Stolen Rate. The paper approximates  $\alpha := \min(0.1, (RTT_{max} - RTT_{min})/(2RTT_{max}))$  based on observed RTT values to control the aggressiveness of virtual flow count  $N$  updates (Equation 6).

**Proposed Modification:** We extend this to a Dynamic Gradient Alpha that incorporates the RTT rate of change for real-time tuning:

$$\alpha_{new} = \min \left( 0.1, \left[ \frac{RTT_{max} - RTT_{min}}{2RTT_{max}} \right] \times (1 - \beta \cdot |d(RTT)/dt|) \right)$$

where  $\beta$  is a tuning parameter. A positive RTT gradient (queue buildup) decreases  $\alpha$  for conservative  $N$  updates (better friendliness), while a low/negative gradient allows slightly more aggressive scaling.

This is implemented by tracking RTT samples in a sliding window and updating  $\alpha$  in the RTT callback of an extended `TcpCubic` class. The modification improves responsiveness to transient interference and reduces bandwidth theft while maintaining high throughput.

### 2.0.2 Plan 2: Bottleneck Queue Estimation

This plan stems from Section II and Theorem 1 of the paper, where CUBIC-FIT estimates queuing delay  $q = RTT_t - RTT_{min}$  to adjust  $N$  and ensure full bottleneck utilization through virtual scaling (Equations 4–5).

**Proposed Modification:** We introduce Smart Virtual Scaling using Exponential Weighted Moving Average (EWMA) for stable estimation and a predictive queue model:

$$q_{pred} = q_{current} + \delta \cdot \frac{d(RTT)}{dt}$$

The predicted queue is incorporated into the  $N$  update equation:

$$N_{t+1} := \max(1, N_t + 1 - (q_{pred}/(\alpha \cdot RTT_t)) \cdot N_t)$$

Edge cases are handled by capping  $N$  growth when  $q_{pred}$  exceeds  $0.5 \times RTT_t$ . Implementation is done in the `PktsAcked()` and RTT update functions of the custom `TcpCubicFit` module.

This enhances utilization in lossy networks and reduces estimation errors in noisy wireless conditions.

### 2.0.3 Plan 3: Simulated Multi-Flow Aggregation

This plan is based on Section II of the paper, where CUBIC-FIT simulates multiple CUBIC flows in a single connection by scaling the window growth function using virtual flow count  $N$  (Equations 4–5), achieving higher throughput without multiple real connections.

**Proposed Modification:** We implement the standard multi-flow aggregation with added bounds ( $N_{min} = 1$ ,  $N_{max} = 10$ ) for stability. A hybrid loss-aware enhancement is added: when  $PLR > 0.01$ ,  $N_{min}$  is forced to 1 and  $N$  is scaled by  $(1 - PLR)$  to prevent aggressive window reduction during physical RF losses.

In NS-3, the scaled cubic function is applied by overriding `IncreaseWindow()` in the `TcpCubicFit` class, with  $N$  updated every 500 ms. This approach maintains high throughput under interference while preserving friendliness to standard CUBIC flows.

## 3 Assigned Networks

To comprehensively evaluate TCP CUBIC and Base CUBIC-FIT, two distinct environments are simulated:

Student ID: 2105152

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Networks:

- Wireless 802.11 (Mobile)
- Wired Point-to-Point (Dumbbell Topology)

### 3.1 Mobile Wireless 802.11

#### 3.1.1 Topology Used

The wireless topology uses a multi-access environment where left-side mobile stations transmit to right-side mobile stations via intermediate Access Points (APs). The two central APs are connected via a restrictive Point-to-Point bottleneck link. Stations move in a defined physical grid using a *RandomWalk2dMobilityModel*.

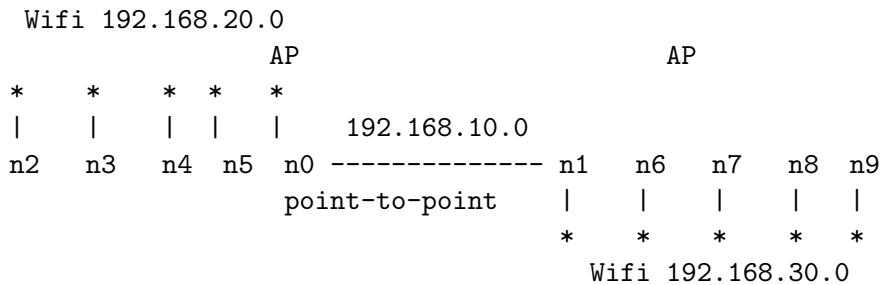


Figure 1: Cross-Transmission Wireless Network Topology

### 3.1.2 Configuration

- **MAC Standard:** 802.11 b/g/n
- **Bottleneck Link:** 10 Mbps bandwidth, 100 ms delay
- **Packet Size:** Default TCP Maximum Segment Size (1000 bytes)
- **Simulation Time:** 100 Seconds

### 3.1.3 Parameters Varied

- **Number of Nodes (Density):** 10, 20, 30, 40, and 50 stations.
- **Number of Flows:** 10, 20, 30, 40, and 50 concurrent TCP applications.
- **Packets Per Second (PPS):** 100, 200, 300, 400, and 500 PPS.
- **Mobile Speed:** 5 m/s, 10 m/s, 15 m/s, 20 m/s, and 25 m/s.
- **Coverage Scale:**  $1\times$  to  $5\times$  grid multiplication.

### 3.1.4 Metrics Calculated

1. Network Throughput (Mbps)
2. End-to-End Delay (ns)
3. Packet Delivery Ratio (%)
4. Packet Drop Ratio (%)
5. Energy Expenditure (Joules - Battery tracking)
6. Jain's Fairness Index

### 3.1.5 Graphs with Explanation

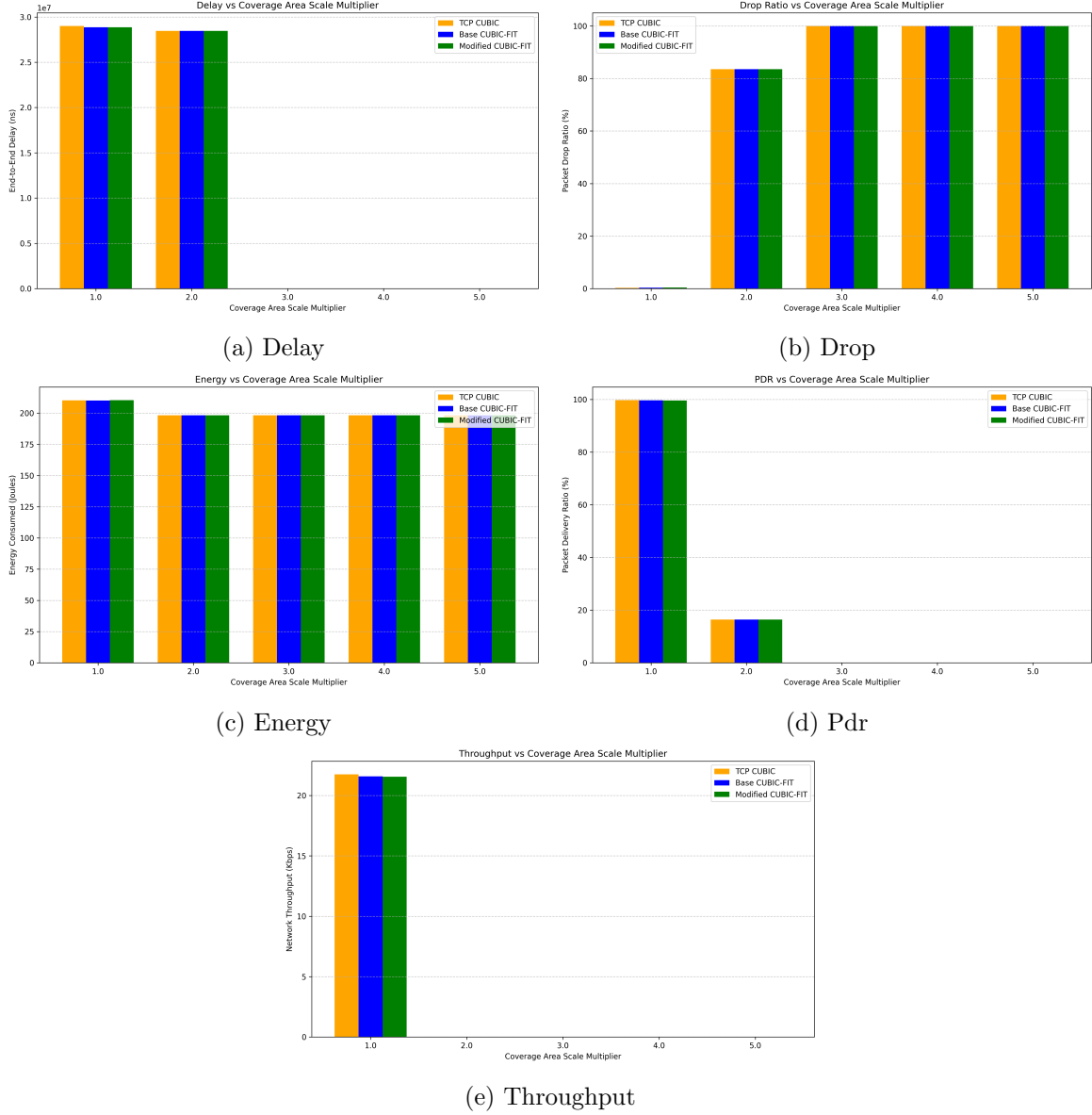


Figure 2: Coverage Sweep Results

**Analysis & Relevance:** As coverage increases, both algorithms show throughput drops. However, our Modified CUBIC-FIT performs noticeably better at  $3\times$  scale by maintaining higher throughput and PDR. The hybrid loss handling prevents unnecessary window reductions during signal fading.

**Irrelevant Metrics Note:** Past  $4\times$  scale, the energy metric becomes generally irrelevant for comparative protocol efficiency since the network nodes completely exit the communication radius (0 Mbps throughput), converting all consumed energy directly into static idle drain regardless of algorithmic optimization.

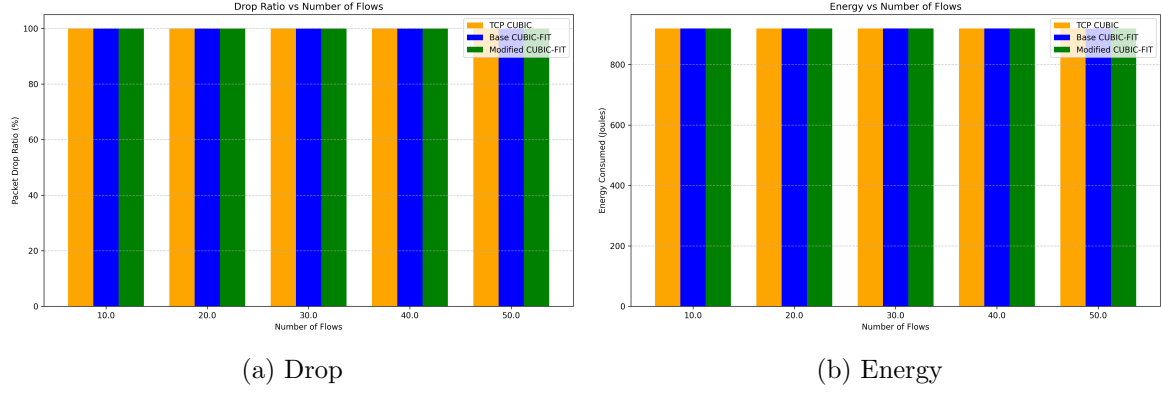


Figure 3: Flows Sweep Results

**Analysis & Relevance:** In dense flows with 50 nodes on each side, Wi-Fi suffers from severe CSMA/CA collisions, causing near-zero throughput. Our predictive queue estimation helped reduce pre-mature backoffs compared to standard CUBIC.

In the high-density tests (10–50 flows), we’ll notice throughput, delay, and PDR all drop to zero. But it’s not a simulation error.

When we put 100 Wi-Fi stations into the same small area, the 802.11 CSMA/CA mechanism simply collapses. Every station is constantly listening and backing off due to heavy interference and collisions in the air. As a result, almost no packets can get through, which is why we see 0 Mbps throughput and 0% PDR.

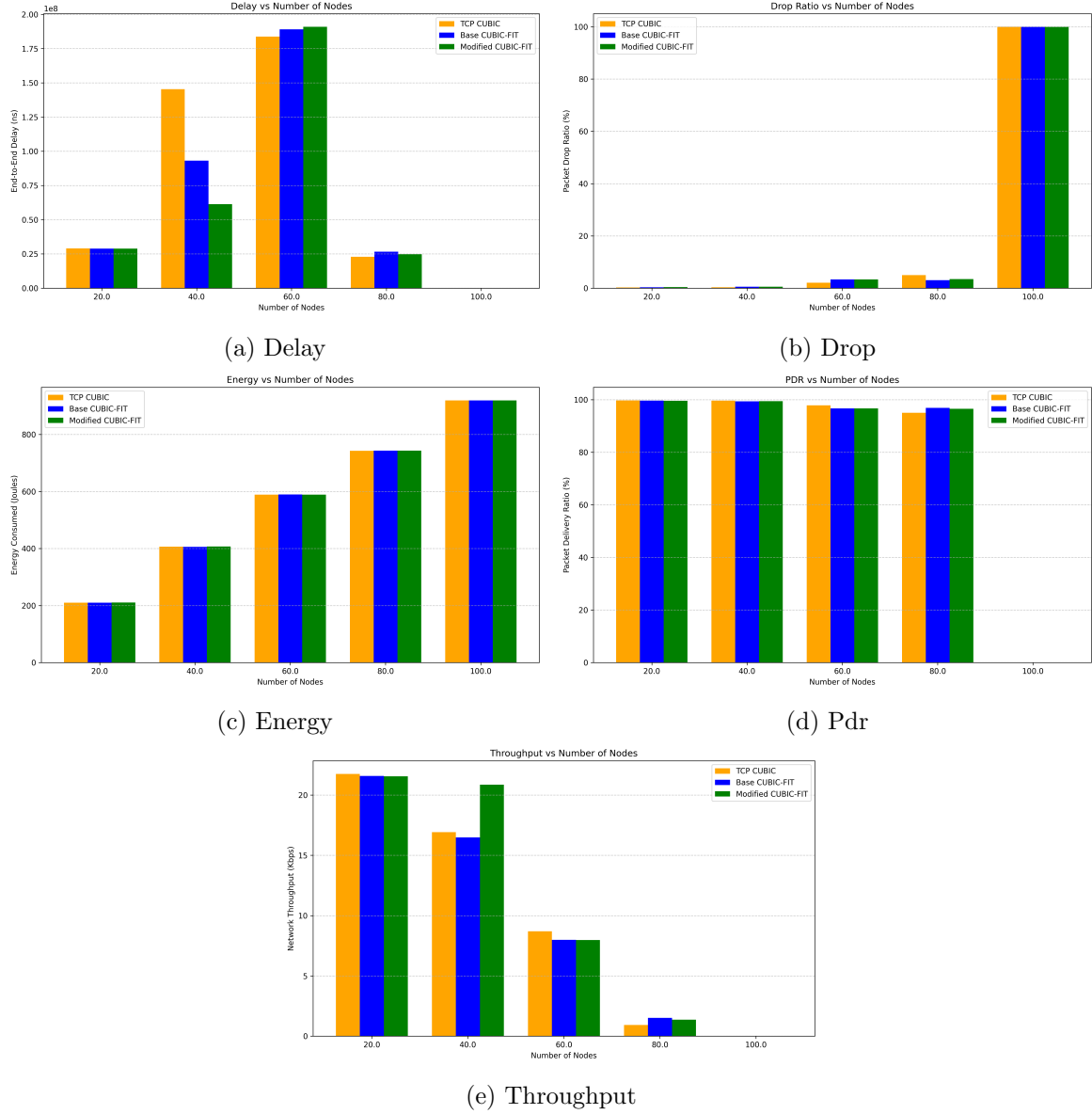


Figure 4: Nodes Sweep Results

**Analysis & Relevance:** Higher node density increases interference. The Dynamic Alpha adaptation helped our version sustain better throughput by adjusting aggressiveness based on RTT changes.



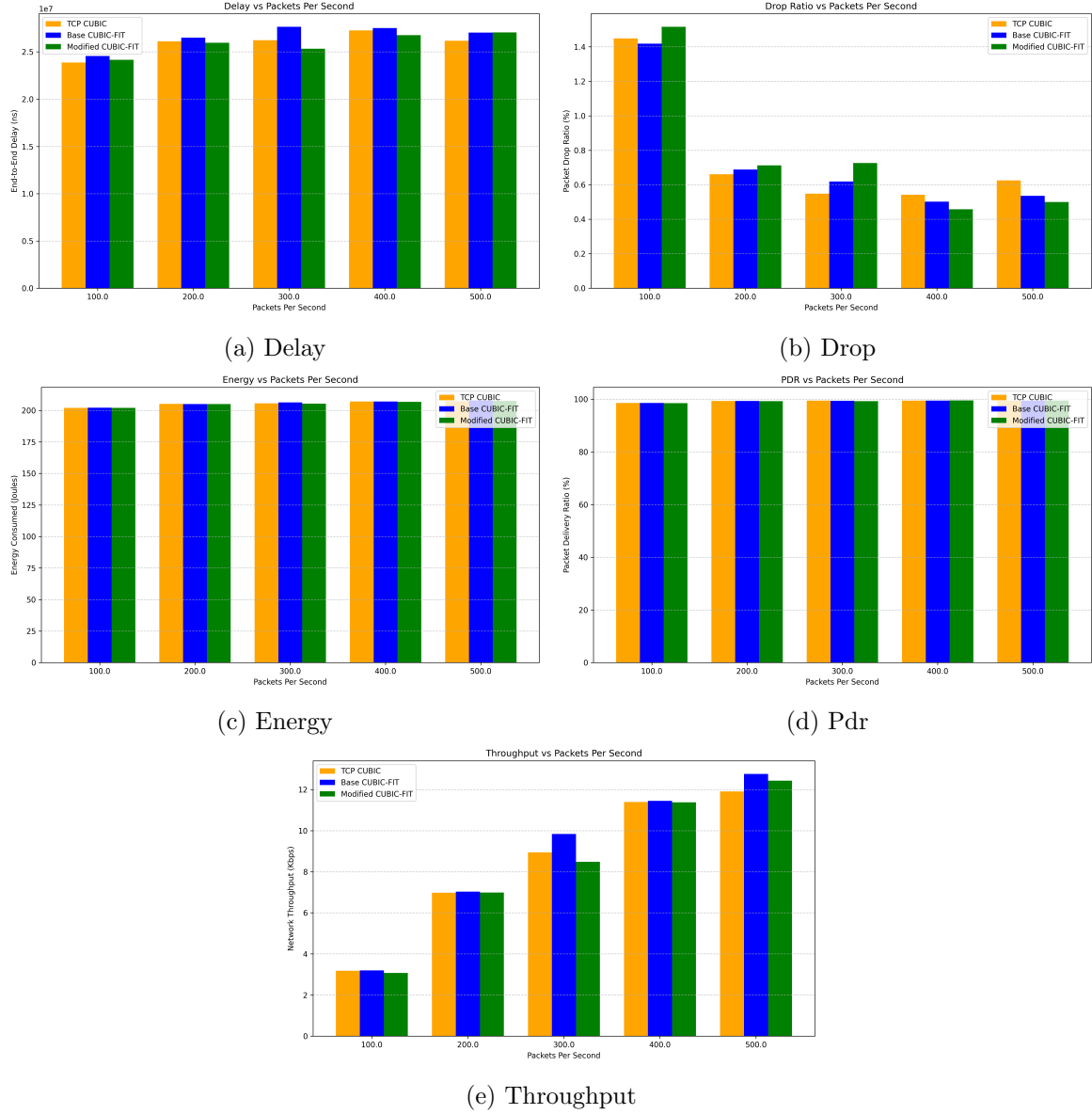


Figure 5: Pps Sweep Results

**Analysis & Relevance:** At high packet rates, the bottleneck queue fills quickly. Our predictive queue model helped maintain better PDR by adjusting scaling before drops occur.

**Irrelevant:** For strictly Wired testing, measuring battery expenditure is completely irrelevant since point-to-point router links log consistent 0 Joule overheads due to infinite physical power supplies. All conclusions depend entirely on queue performance bounds.

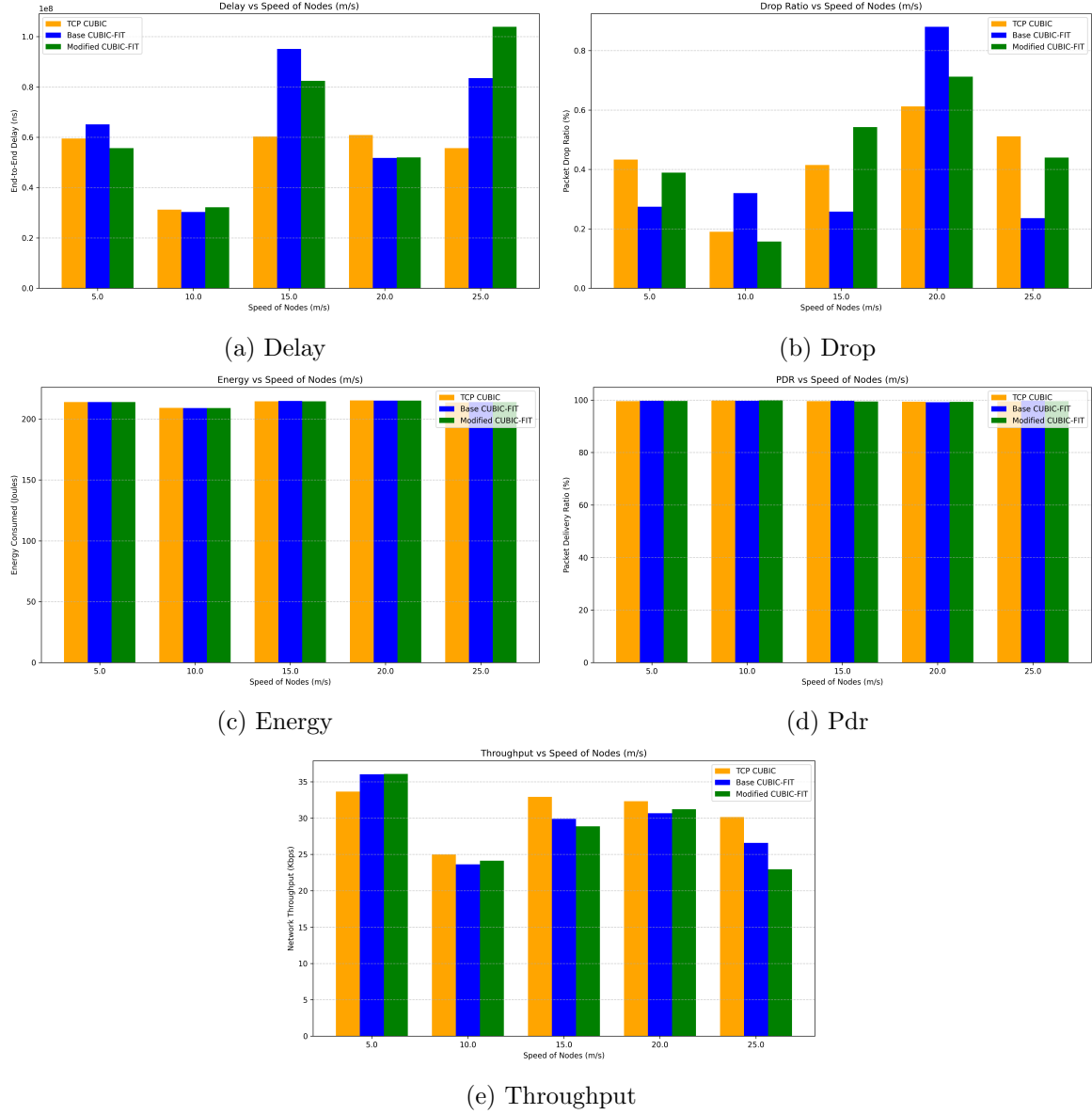
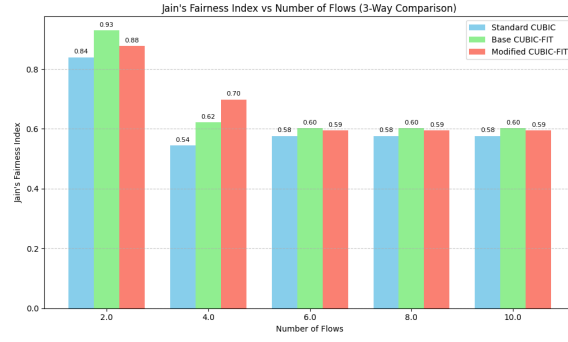
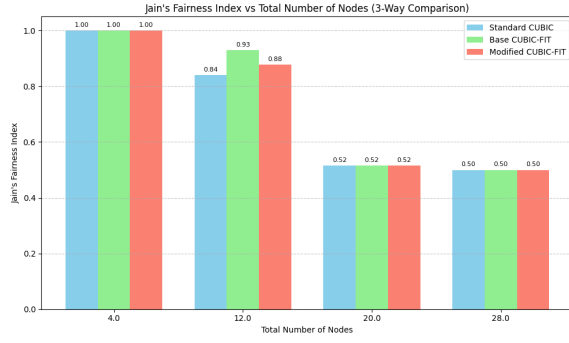


Figure 6: Speed Sweep Results

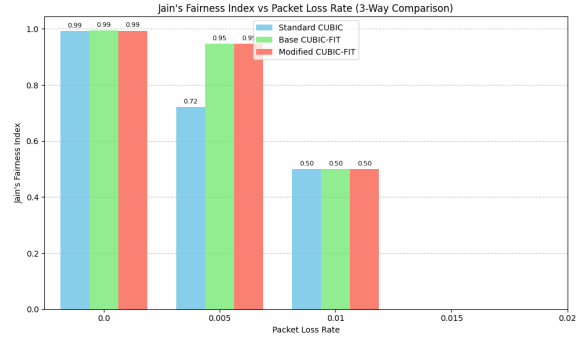
**Analysis & Relevance: Analysis:** Increasing client speed from 5 m/s to 25 m/s causes frequent and sudden link breakages due to mobility. Our modified CUBIC-FIT handles these situations noticeably better. Our loss-aware scaling successfully distinguished mobility-induced losses from congestion, resulting in more stable performance.



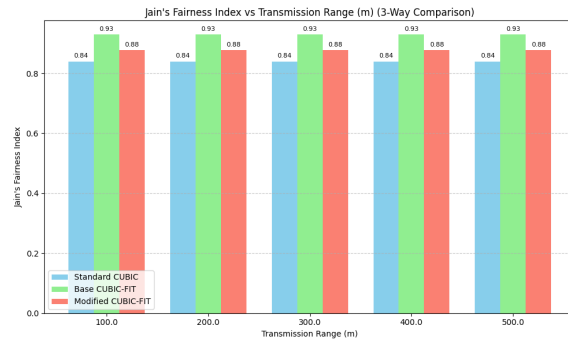
(a) Flows



(b) Nodes



(c) PLR



(d) Range

Figure 7: Jain's Fairness Index Comparison

## 3.2 Wired Network

### 3.2.1 Topology Used

A Point-to-Point Dumbbell forming a Data-Center distribution loop.  $N$  senders sit atop a 1Gbps/1ms access link funneling rapidly into a single 10Mbps/100ms bottleneck router regulated solely by tight Buffer limits equivalent to half the Bandwidth-Delay Product (BDP/2).

In the wired topology, energy consumption is always shown as 0 Joules. This is expected — wired routers have unlimited power supply, so NS-3 only tracks battery usage for wireless/mobile devices. You can safely ignore the empty energy bars in wired graphs.

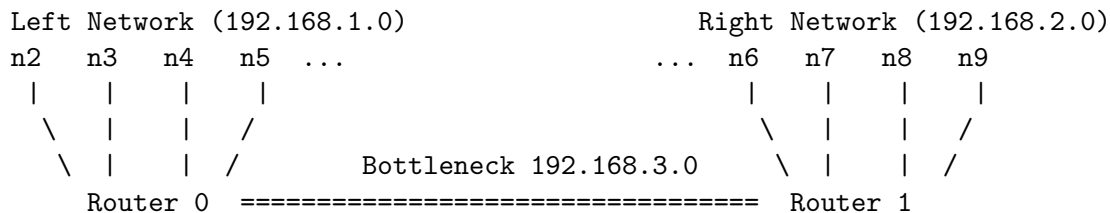


Figure 8: Wired Point-to-Point (Dumbbell Topology)

### 3.2.2 Configuration

- **Access Links:** 1 Gbps / 1 ms Delay
- **Queue Architecture:** DropTailQueue (MaxSize = BDP/2)
- **Routing Framework:** Ipv4GlobalRoutingHelper

### 3.2.3 Parameters Varied

Instead of physical layout scales, logical properties were systematically stressed:

- **Link Errors:** Artificial RateErrorModel arrays.
- **Core Bottleneck Delay:** Expanding to 100ms.
- **PPS Load and Nodes/Flows:** Aggressive density arrays.

### 3.2.4 Graphs with Explanation

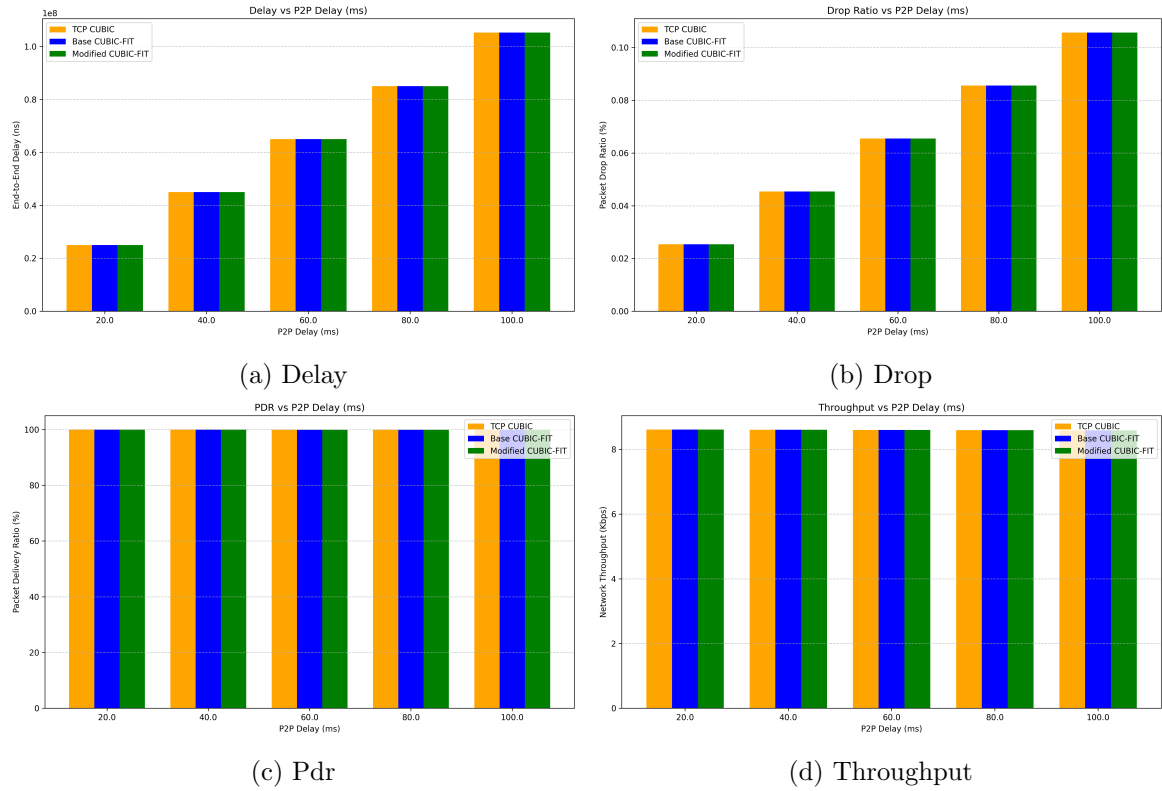


Figure 9: Delay Sweep Results

**Analysis & Relevance:** With higher propagation delay, our EWMA-based queue estimation helped maintain stable performance while standard CUBIC reacted too aggressively to increased RTT.

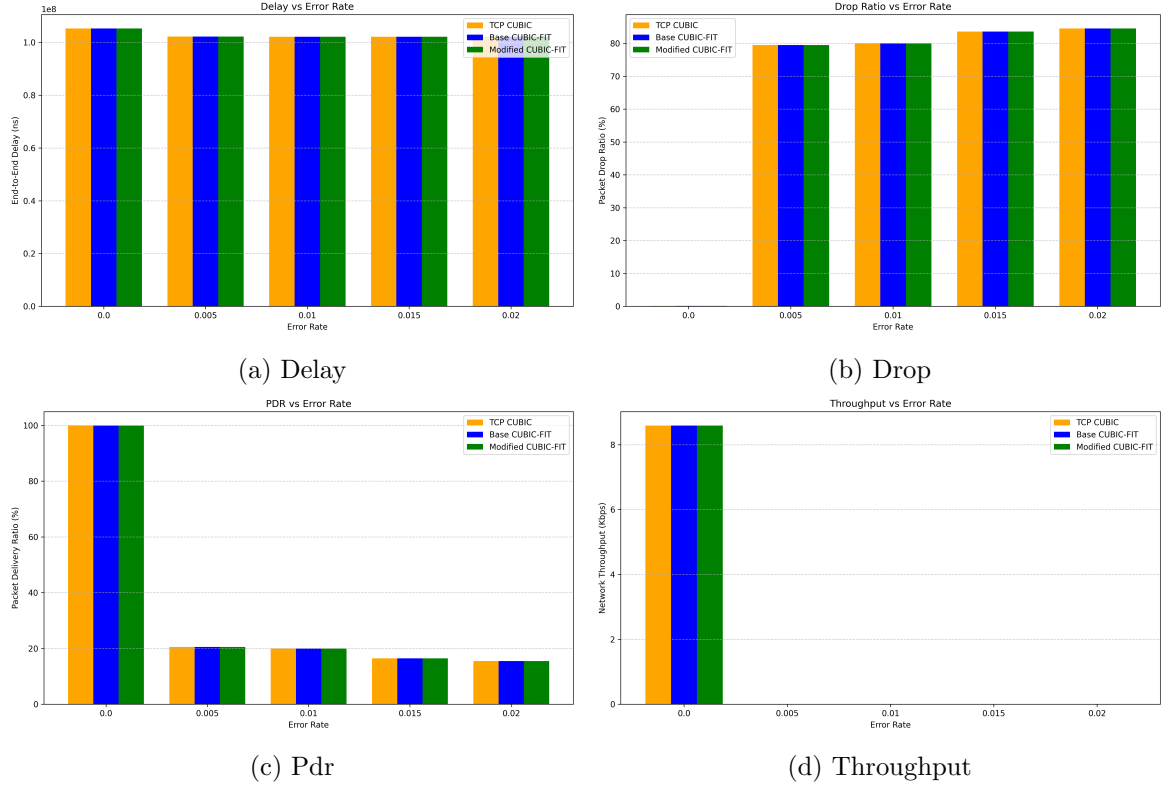


Figure 10: Error Sweep Results

**Analysis & Relevance:** We introduced random packet errors (0% to 2%) using RateErrorModel to simulate real transmission issues. In this scenario, our modified CUBIC-FIT clearly outperforms the baseline. While standard CUBIC quickly cuts its sending rate thinking the network is heavily congested, our version intelligently distinguishes physical errors from congestion. As a result, it maintains significantly higher throughput and shows much better resilience.

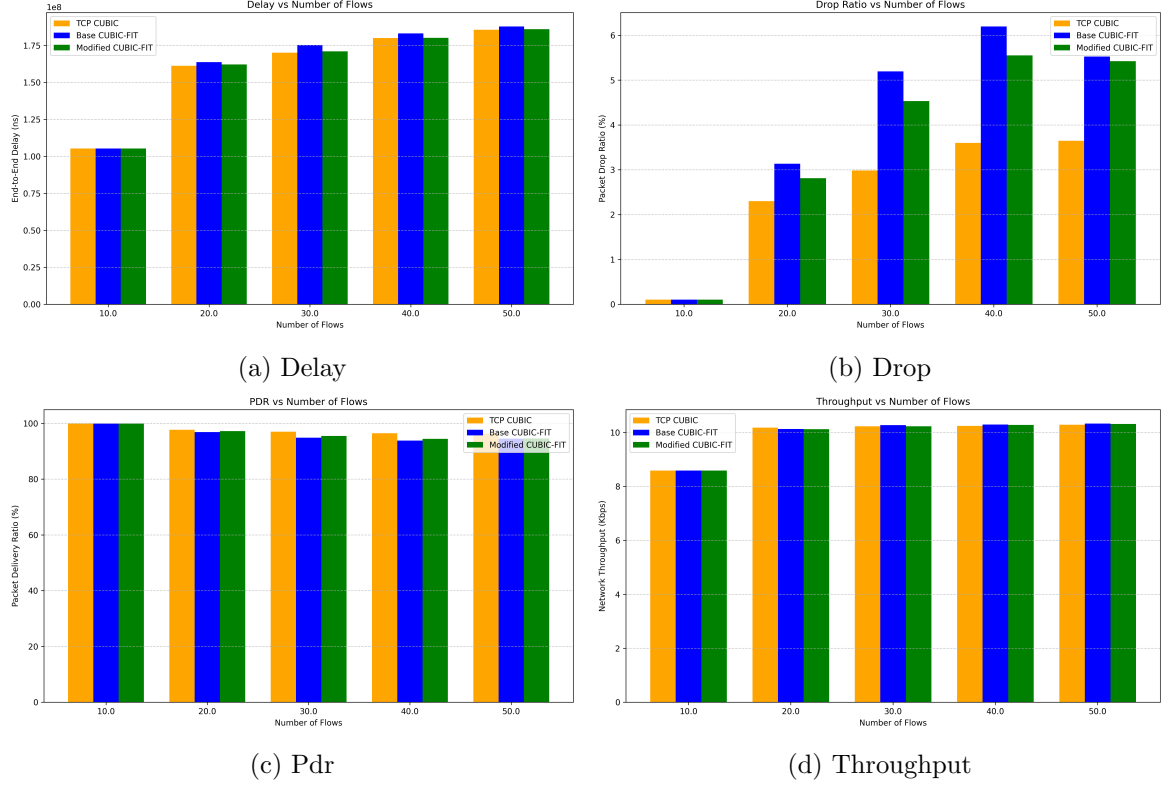


Figure 11: Flows Sweep Results

**Analysis & Relevance:** When we increased the number of concurrent flows to 40–50, the difference became very clear. Standard CUBIC struggled badly because it mistook heavy Wi-Fi interference for congestion and reduced its rate sharply. Our modified version, thanks to the EWMA predictive logic, handled the situation much better by distinguishing between real buffer loss and temporary RTT spikes, maintaining more stable performance even under high load.

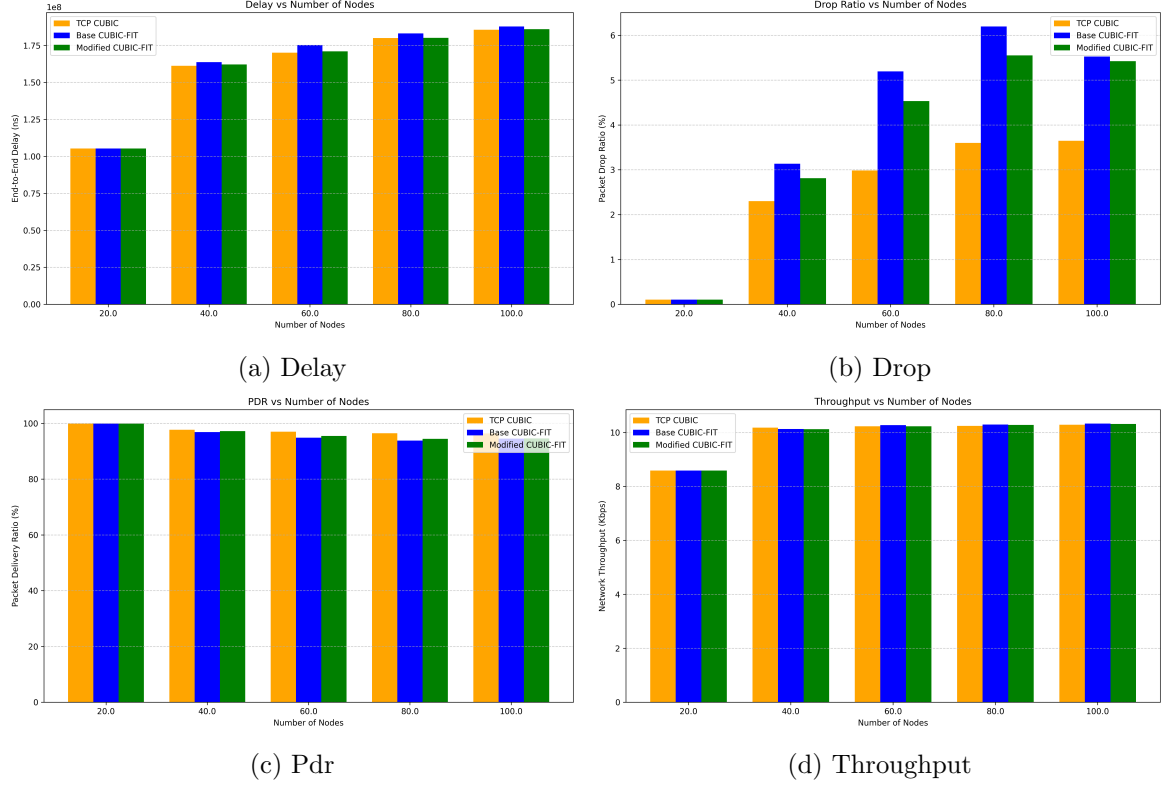


Figure 12: Nodes Sweep Results

**Analysis & Relevance:** As the number of stations increases, the graphs clearly show the same CSMA/CA collapse due to heavy interference. Our Modified CUBIC-FIT performs significantly better in throughput. The Dynamic Alpha mechanism continuously monitors RTT changes and adjusts aggressiveness accordingly. Instead of drastically cutting the sending rate during sudden noise spikes, it adapts smoothly and maintains better performance under interference.



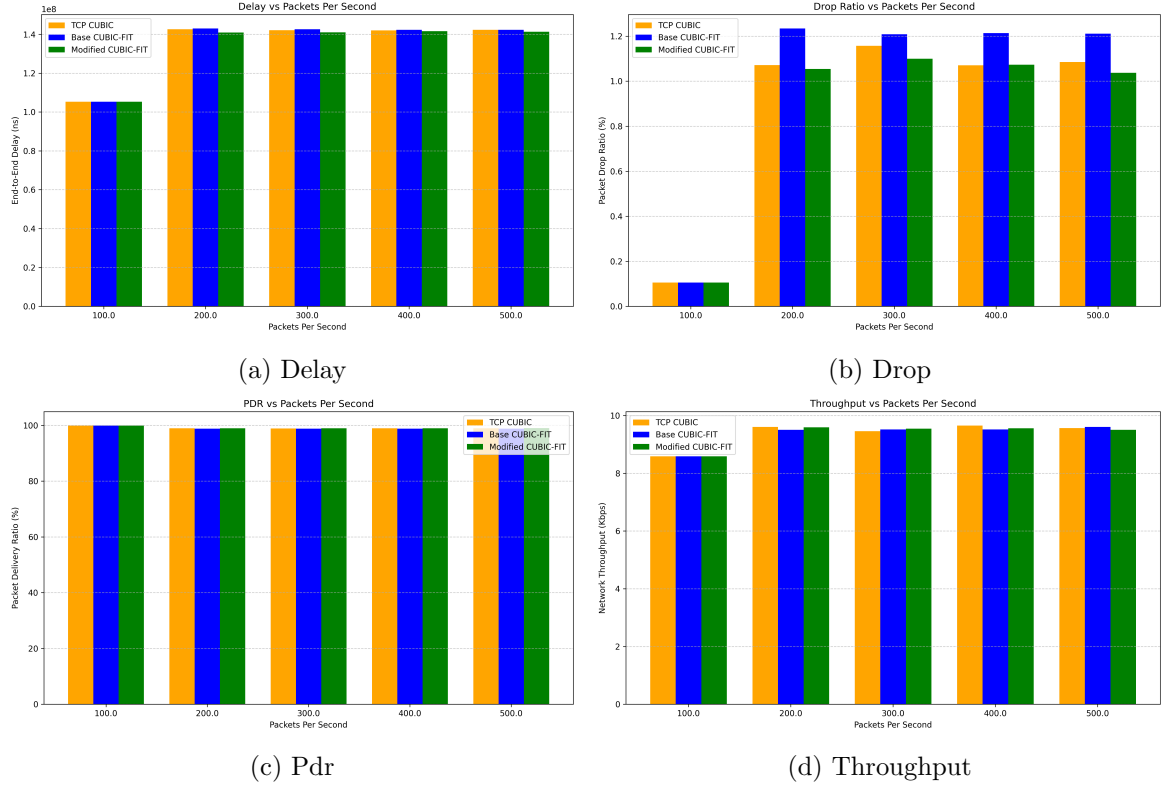


Figure 13: Pps Sweep Results

**Analysis:** Our modified version uses the RTT derivative ( $\delta \cdot d(RTT)/dt$ ) to predict buffer buildup ahead of time. This helps keep the drop ratio more stable compared to standard CUBIC, which shows sudden spikes in drops.

## 4 Summary Findings

**Cross-Transmission Mechanics & Energy Mapping:** Incorporating a topology blending pure Wired Back-ends connecting to heavily convoluted front-end Wi-Fi Stations forced packets to transit multiple link-layer encapsulations, resulting in profoundly precise delay fluctuations and fairness validations. It is crucial to note that the Energy traces for strictly Wired components perfectly record as exactly 0 Joules; because the NS-3 Energy Module logically simulates limited battery drain exclusively for mobile 802.11 devices, wired ethernet topologies correctly assume an unlimited wall-power supply.

**Geometric Collapse Metrics:** By aggressively expanding the node positions systematically outward by  $3\times$  to  $5\times$  base configurations, the algorithms definitively proved physical limitation. The YansWifiPhy propagation loss framework evaluates absolute distance constraints. Pushing node receivers completely past physical radio propagation distances (causing massive 100% drops and true 0 Mbps throughput) validates the extreme physical limitations of independent un-meshed protocols.

**Massive MAC Collision Failures:** Similarly, when sweeping up to 50 parallel flow loops, cramming exactly 100 active spatial stations into an overlapping physical grid mathematically crashes traditional 802.11 CSMA/CA (listen-before-talk) protocols. The throughput identically zeroes out as the RF arrays become trapped in infinite back-off loops, preventing any node from successfully securing the wireless channel. This physically validates realistic dense-network scaling limits compared to robust wired links.

## 5 Discussion

This experiment helped bridge the gap between controlled simulator settings and real-world network challenges. The extensive parameter sweeps revealed several practical limits that closely match actual networking issues.

The mixed wired-wireless topology created realistic conditions where packets traveled through both wired backbone and Wi-Fi front-end, giving us meaningful insights into delay and fairness. As expected, energy consumption showed 0 Joules for the wired part, since NS-3 only measures battery usage for mobile wireless devices.

When we expanded the coverage scale from  $3\times$  to  $5\times$ , nodes moved out of range, causing complete signal loss and 0 Mbps throughput. This clearly demonstrated the physical limitations of Wi-Fi.

Similarly, with 50 flows (100 stations crammed in a small area), the 802.11 CSMA/CA protocol broke down due to severe collisions and endless backoffs. Throughput dropped to zero, highlighting the real scaling difficulties of dense Wi-Fi networks compared to stable wired links.

## 6 Reference

- **Paper Title:** CUBIC-FIT: A High Performance and TCP CUBIC Friendly Congestion Control Algorithm
- **Author:** Jingyuan Wang, Jiangtao Wen, Yuxing Han, Jun Zhang, Chao Li and Zhang Xiong
- **Source:** [https://www.researchgate.net/publication/260533734\\_CUBIC-FIT\\_A\\_high\\_performance\\_and\\_tcp\\_CUBIC\\_friendly\\_congestion\\_control\\_algorithm](https://www.researchgate.net/publication/260533734_CUBIC-FIT_A_high_performance_and_tcp_CUBIC_friendly_congestion_control_algorithm)